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safeExam.Ai — Title Page

Next-Generation AI Proctoring & Automated Online Examination System

Core AI Engine: Agent ELI powered by Custom Trained YOLOv11 Model

Presented by: Department of Computer Science & Engineering

Architecture: 1 FPS Edge Computer Vision • Multi-Tier Browser Security • Zero-Streaming Bandwidth

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Abstract

- **Problem:** Traditional online assessments suffer from widespread cheating (phones, notes, tab switches, reloads) and high invigilation costs.
- **Proposed Solution:** safeExam.Ai provides edge-based computer vision (YOLOv11) combined with browser-level security guards.
- **Key Features:** 10-demerit points ceiling, instant reload auto-submit, 5-minute offline network hold, and live teacher surveillance tabs.
- **Impact:** Eliminates costly video streaming bandwidth (<5 kbps/user) while ensuring instant sub-35ms violation detection.

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Introduction & Problem Motivation

- Rapid shift to e-learning and remote certifications requires verifiable academic integrity.
- Human proctors cannot reliably monitor 50+ webcam feeds simultaneously without fatigue and oversight.
- Heavy cloud-based proctoring software suffers from severe video bandwidth bottlenecks and student privacy invasions.
- safeExam.Ai solves this via decentralized, on-device AI vision and automated progressive intervention.

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Problem Statement

- **Secondary Devices:** Students looking at smartphones or second laptops off-camera.
- **Unauthorized Notes:** Open physical textbooks and cheat sheets placed near keyboards.
- **Browser Collusion:** Copy-pasting exam questions to AI tools (ChatGPT) via shortcuts (Ctrl+C, Ctrl+V).
- **Tab Switching & Window Navigation:** Toggling external search tabs or background messaging software.
- **Network Sabotage & Page Reloads:** Deliberately disconnecting Wi-Fi or reloading pages to bypass timers.

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Aim and Objectives

- **Aim:** Design, implement, and benchmark an automated, zero-latency online exam and proctoring platform.
- **Objective 1:** Train and deploy an edge YOLOv11 model for real-time device, book, and presence detection.
- **Objective 2:** Implement deterministic browser guards (blocking shortcuts, context menu, and reload attempts).
- **Objective 3:** Engineer a resilient 5-minute network offline hold and automated fail-safe submission.
- **Objective 4:** Deliver an interactive teacher surveillance console with per-student tab cards and live audit logs.

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Project Scope

- **In-Scope:** Web-based examination interface (MCQ & Written), 1 FPS YOLOv11 edge vision, progressive warning modals, 10-demerit limit, 5-minute offline hold, and teacher surveillance dashboard.
- **Out-of-Scope:** Proprietary OS kernel drivers (focusing on universal modern web standards) and hardware biometric fingerprint scanners.

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Literature Review & Benchmark Analysis

- **Safe Exam Browser (SEB):** Rigid OS desktop install, zero object detection, prone to crashes.
- **Proctorio / Honorlock:** Expensive per-seat licensing, heavy video uploads (1.5 Mbps/user), privacy concerns.
- **Mettl / Mercer:** High server costs, requires post-exam human reviewers, delayed results.
- **safeExam.Ai:** Custom YOLOv11 edge AI, ultra-low bandwidth (<5 kbps metadata), instant progressive penalties, 10-demerit limit.

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Proposed System Overview

- **Agent ELI Vision Core:** Custom-trained YOLOv11 model detects secondary devices, books, and absence on the edge (<35ms inference).
- **Progressive Integrity Protocol:** 1st shortcut warns candidate (0 pts); 2nd attempt onwards adds +1 demerit. Cumulative 10 demerits trigger auto-submit.
- **Network & Reload Shield:** Page reload (F5 / Ctrl+R) triggers instant auto-submit. Disconnects initiate a 5-minute unclosable hold modal.

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System Requirements (Hardware & Software)

- **Client Device:** Dual-core 2.0 GHz+ CPU, 4 GB RAM, standard 720p HD webcam, 1 Mbps connection.
- **Client Software:** Modern web browsers (Chrome, Edge, Brave, Firefox) with WebRTC & WebAssembly.
- **Backend Server:** Node.js v20+, Express.js REST API Gateway, MongoDB Atlas document database.
- **AI Inference:** Custom Trained YOLOv11 Object Detection Engine running on edge client.

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System Methodology & Development Lifecycle

- **Phase 1:** Custom YOLOv11 dataset curation and edge optimization for proctoring environments.
- **Phase 2:** Heuristic browser security engineering (keydown, clipboard, beforeunload, visibility API).
- **Phase 3:** Resilient state engineering (10s heartbeats, 30s auto-save, 5-min network hold modal).
- **Phase 4:** Teacher command console development (per-student tab cards, live incident feed, CSV export).
- **Phase 5:** Comprehensive stress testing and verification.

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System Architecture (3-Tier Model)

- **Presentation & Edge AI Tier:** React 18 SPA, WebRTC webcam capture, YOLOv11 edge inference, browser traps.
- **Application Tier:** Node.js & Express.js REST Gateway, JWT & RBAC middleware, Proctoring Controller, 10s heartbeat scanner.
- **Data & Persistence Tier:** MongoDB Atlas collections (Exams, ExamQuestions, Students, StudentExams, ProctoringLogs).

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Use Case Model

- **Student:** Login, verify passcode, answer questions (MCQ/Written), observe live demerit meter, receive warning modals, submit attempt.
- **Teacher:** Create/reorder exams, monitor live student roster tabs, inspect filtered incident feeds, grade written answers, download CSV audits.
- **Agent ELI:** Edge vision loop, trap reloads/shortcuts, enforce 5-min network hold, accumulate demerit points, trigger auto-submit.

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System Workflow & Activity Lifecycle

- Start Exam -> Webcam & YOLOv11 Initialization -> Active Proctoring Loop.
- Prohibited Shortcut: 1st warning modal -> 2nd time +1 demerit.
- Page Reload (F5/Ctrl+R): Instant Auto-Submit via Beacon API.
- Tab Switch >10s: 5s countdown modal -> Auto-Submit.
- Phone Detected >=5s: 5s confirmation modal -> Auto-Submit if unconfirmed.
- Network Disconnected: Unclosable 5-min hold modal -> Auto-Submit at 5:00.
- Reaching 10 Demerits: Immediate Auto-Submission.

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Database Design & ER Model

- **Student & Teacher:** User identities, roles, department, credentials.
- **Exam & ExamQuestion:** Title, duration, pass marks, question type (MCQ/Written), options, sort order.
- **StudentExam (Attempt):** Status, demerit points, start/end timestamps, last_heartbeat date.
- **ProctoringLog (Audit):** Granular event log with activity_type, details, demerits, student_id, and timestamp.

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Technology Stack Deep-Dive

- **Frontend:** React 18, Vite, Tailwind CSS, Lucide React, HTML5 Canvas API.
- **AI Computer Vision:** Custom Trained YOLOv11 Object Detection Engine (Agent ELI).
- **Backend:** Node.js v20+, Express.js REST Gateway, JWT, BCrypt, CORS.
- **Database:** MongoDB Atlas, Mongoose ODM, MongoDB Aggregation Pipelines.

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System Implementation Details

- **ExamInterface.jsx:** Central student exam view handling camera, question answering, and modals.
- **TeacherProctoring.jsx:** Real-time surveillance view with student roster tab buttons and filtered incident feed.
- **proctorController.js:** Atomic demerit scoring and threshold enforcement (Max 10 pts).
- **studentController.js & teacherController.js:** Heartbeat receiver, 5-minute offline scanner, hybrid grading.

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Key Features & Proctoring Enforcement Matrix

1. Prohibited Shortcuts & Clicks: 1st warning modal (0 pts); 2nd time onwards +1 Demerit.
2. Page Reload: Instant Auto-Submit on F5, Ctrl+R, or refresh button.
3. Tab Switch >10s: +2 Demerits and 5s countdown auto-submit modal.
4. Network Disconnected: Exam on hold (no answering); auto-submits if offline >5 minutes.
5. Mobile Phone Detected: Phone >=5s triggers 5s confirmation modal -> Auto-Submit if unconfirmed.
6. Notebook / Notes: +1 Demerit point added for every 5s visible.
7. 10 Demerit Points Limit: Automated ceiling triggers immediate auto-submission.

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User Interface Architecture & Highlights

- **Student Workspace:** Clean questions layout, camera preview with bounding boxes, live demerit meter (0-10), local terminal feed.
- **Warning Modals:** Shortcut modal, tab switch modal, phone confirmation modal, 5-min network hold modal.
- **Teacher Console:** Metric cards, student roster tab cards (In Exam, Submitted, Offline), filtered incident table with color badges.

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Testing & Quality Assurance

- TC-01: Prohibited Shortcuts (Ctrl+C/V/A) -> Trapped, 1st warning modal, 2nd time +1 pt [PASS]
- TC-02: Page Reload (F5/Ctrl+R) -> Immediate auto-submit triggered [PASS]
- TC-03: Tab Switch >10s -> 5s countdown modal fired, auto-submitted [PASS]
- TC-04: YOLOv11 Phone Detection -> Warning modal with 5s countdown prompted [PASS]
- TC-05: Network Disconnection -> Unclosable Hold modal displayed; auto-submitted at 5:00 [PASS]
- TC-06: 10 Demerits Limit -> Immediate auto-submission executed at 10 points [PASS]

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Results & Discussion

- **Zero Video Bandwidth Overhead:** Consumes <5 kbps per candidate (metadata only).
- **YOLOv11 Edge Speed:** Achieves 25-35 ms execution time per frame on standard laptops.
- **Student Privacy:** Webcam stream is analyzed in local device memory and never uploaded.
- **Cheating Deterrence:** Progressive demerits and unclosable hold modals eliminate loopholes.

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Current System Limitations

- **Webcam Field of View:** Standard laptop cameras capture a frontal angle; blind spots exist beneath the desk.
- **Lighting Sensitivity:** Extreme low light can reduce object confidence scores.
- **Audio Eavesdropping:** Current release focuses strictly on visual and browser events.

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Future Roadmap & Enhancements

- **Acoustic Speech AI:** Implement browser voice detection to flag whispering or background speech.
- **Dual-Camera Setup:** Pair student smartphones as secondary side-angle perspective cameras via WebRTC.
- **Gaze & Head Pose Tracking:** Facial landmark estimation to detect prolonged off-screen looking.
- **LLM Auto-Grader:** Generative AI for automated scoring of written essay questions.

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Conclusion

- safeExam.Ai successfully bridges the gap between remote exam access and academic integrity.
- Custom YOLOv11 edge vision combined with browser guards delivers proactive cheating deterrence.
- Eliminates cloud video streaming costs while providing transparent live surveillance for educators.
- Production-ready architecture verified with zero build errors.

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References

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Q&A; / Thank You

THANK YOU!

safeExam.Ai — Integrity Powered by AI Vision

Open for Questions & Discussion.